

Immersive and Enjoyable Explanations - On Distinct Explainability Requirements in Games

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Abstract. [Context and motivation] When complex systems are difficult to understand and interact with, explanations have proven effective in supporting users and enhancing user experience (UX). Explainability research has extensively examined UX factors critical to productivity software, such as trust and learnability. [Question/problem] However, it has largely overlooked key UX factors of entertainment software, including immersion, enjoyment, and emotional engagement. Video games provide a compelling context for investigating this research gap, as they combine complex interactive systems with rich emotional and immersive experiences. Since there is no general understanding of players as the stakeholders of software explanations, two central questions remain: (1) what types of explanations do players need, and (2) how do explanations influence players' user experience? [Principal ideas/results] In this work, we report on two complementary studies: a survey and a controlled experiment. The survey assessed participants' general need for explanations in digital games. The experiment involved participants playing a video game, *Ori and the Blind Forest*, either with or without explanations, and evaluating their user experience. [Contribution] Our findings provide two key contributions: (1) we identify explanation needs that are unique to video games and distinct from other software domains, and (2) we offer empirical evidence that explanations may enhance players' user experience, improving immersion and enjoyment.

Keywords: Explainability · Requirements Engineering · User Experience
· Entertainment Software · Digital Games

1 Introduction

Explanations are an essential part of many complex software systems [21, 22]. The respective non-functional requirement explainability describes the ability of a software to explain itself, its behavior, and its interactions [12, 22, 39]. In recent years, explainability has gained major attention in the field of artificial intelligence (AI) [1, 39]. In particular, explanations have been used to make AI behavior and

decisions more transparent and understandable to human users [12,13,39]. Outside of AI, software engineering research has applied explainability concepts to make system aspects such as security [4,61] and privacy [8,9] more understandable. Meanwhile, user experience research has identified explanations as a suitable means to guide user interactions, improving the learnability and usability of the software [3,20,21].

Video games and their diverse technology and gameplay features may impose various challenges on their players. Game designers walk a fine line between making their games accessible, but also challenging and engaging [41,52]. As such, explanations have long been an essential part of many video games. One example of this are tutorials for gameplay elements that are not immediately intuitive [3,5]. Another example are explanations within the narrative of story-driven games, which aim to construct a coherent in-game logic within the player’s mind [35,42].

Requirements engineering research has concluded that different stakeholders of a software need different explanations [13,22,39]. This is also true for video game research, where different works have already addressed many different explanation types [3,35,38]. However, these works often address explanation needs in games on a case-by-case basis, rather than approaching them under the broader term of explanation requirements. As such, there is no general understanding of players as the stakeholders of software explanations.

In this paper, we examine video games as explainable software, through the lens of requirements engineering. We discuss relevant related work from the fields of requirements engineering and human-computer interaction, and report on two complementary studies: a survey and an experiment. In the survey study, 50 participants gauged their general need for explanations in video games. In the experiment, 20 participants played Moon Studios’ *Ori and the Blind Forest* – either with or without explanations. Using a post-experiment questionnaire, we analyzed their user experience and elicited explanation needs for the game. Our work contributes two major findings: 1) video games invoke a unique prioritization and categorization of explanation needs, which is distinct from other everyday software, and 2) explanations enhance players’ user experience, improving their fun and immersion. These findings underline the importance of integrating explanation design into game development as a formal requirement.

The rest of this paper is structured as follows: Section 2 covers related work. Section 3 details the methodology of our initial research, while Section 4 covers the initial findings. We discuss the implications of our work as well as future research directions in Section 5 and conclude this paper in Section 6.

2 Related Work

2.1 Requirements Engineering for Explainable Systems

Explainability is an emerging non-functional requirement in complex software systems. It describes the ability of a system to provide explanations for any system aspect, making itself more understandable to its stakeholders [12,22]. With the

emergence of AI, research into explainability has boomed since 2018 [26]. Most research in the field of explainable AI focuses on explaining the inner workings of the system [17, 58]. However, requirements engineering for explainable systems has also been conducted in the context of non-AI software, revealing that end-users require a diverse range of software explanations. These include system aspects such as user interactions [19, 22], privacy information [8, 9], or domain knowledge [21, 48].

Recent works in the field of requirements engineering for explainable software have focused on everyday software systems [22, 59], which video games are considered to be a part of [28]. Crawling user feedback, such as app reviews, has proven particularly useful for automated need assessment [46, 59]. The use of taxonomies that categorize types of explanation needs has proven to be effective in supporting discussions about these needs [21, 22, 59]. When combined with workshops, such taxonomies have proven to help participants to establish user-specific requirements [21, 47]. However, so-called hypothetical bias [30] impedes the elicitation of explanation needs, as users find it difficult to envision what questions they might have without having used the system themselves. Considering this, prototyping [57] or elicitation during system runtime [19] are effective methods for identifying actual explanation needs.

2.2 Explanations and User Experience

User experience includes all interactions of the user with the product, the service, and the responsible company [45]. In particular, the perception and feelings of users toward the system are highlighted [2, 31, 32]. Orzikulova et al. [49] demonstrated that explanations can improve the user experience, especially in terms of effectiveness. This is in line with frequently stated goals of explanations, including trust, user satisfaction, and symbiosis [26]. The human-computer interaction (HCI) community even explicitly lists user experience as a goal of explainability [26]. According to El Ali et al. [25], providing good explanations includes avoiding information overload, enabling personalization, and improving the design of the explanation, i.e., its presentation and placement. Bertrand et al. [7] suggest providing selective, mutable, and dialogic explanations, emphasizing that interactive explanations increase perceived usefulness and performance.

Deters et al. [20] analyzed the influence of explainability on user experience as a whole by examining the influence on each UX factor individually. Their findings show that the influence of explanations on UX factors such as fun, immersion, and emotional engagement has not yet been reviewed by scientific literature. UX experts suspect both positive and negative influences on these factors, revealing an interesting research gap [20]. Entertainment software, including video games, offers a compelling research context to address this gap, as its design focus lies on hedonistic UX factors rather than transparency and productivity [44]. Given that game development heavily relies on non-functional dimensions (e.g., enjoyment and immersion), established requirements engineering practices and frameworks may help to assess the players' perspective [37].

2.3 Requirements Engineering for Digital Games

Kasurinen et al. [37] and Callele et al. [10] highlight the importance of requirements engineering techniques in game development processes, although they are not yet systematically applied. Given that game development tends to focus on non-functional requirements (e.g., fun and immersion), established requirements engineering practices may help to assess the players' perspective [37].

Daneva [16] analyzed gameplay requirements for massively multiplayer online role-playing games (MMORPGs). They identified *playability* as a key non-functional requirement of digital games, which, similar to usability, is closely related to explainability. Whereas a player's explanation needs specifically refer to a deficit of information during gameplay, playability needs refer to their overall capability to engage with the game. This includes explanations, but also other features such as accessibility and usability elements. Therefore, when addressing playability needs, requirements engineers can leverage explanations as a means to improve playability and, by extension, player experience.

2.4 User Experience in Digital Games

Player Experience (PX) refers to user experience in the specific context of digital games [44]. While PX shares its foundational elements with UX, it diverges in critical ways. Traditional UX often prioritizes task efficiency and functional usability, whereas PX centers on hedonic dimensions such as fun, immersion, and emotional engagement [44]. In digital games, the experience itself is the goal, making PX a more affective, enjoyment-centered construct. Game usability plays a central role in shaping PX. As Rajanen and Tapani [54] note, games – like any interactive software – require strong usability to ensure a positive experience. However, usability in games extends beyond traditional productivity software metrics. Depending on the genre, games provide challenge, support learning, and sustain player motivation through dynamic psychological components such as flow, immersion, and emotions [15, 18, 54]. Pedersen et al. [51] highlight challenge and learnability as key to enjoyment, especially in games with procedurally generated content. Their findings indicate that players prefer smooth progression, but also require challenge to achieve flow, underscoring the need to balance difficulty and accessibility in game design [15].

As digital games are marketed directly to individual consumers rather than organizations, a compelling and enjoyable PX directly affects product success and player retention [50]. From a software development perspective, Pagulayan et al. [50] offer a user-centered design framework for games that addresses both evaluation and optimization strategies. They emphasize that games differ from productivity applications in that their primary focus is on the process of interaction rather than outcome achievement. Games construct their own internal goals, and as such, must cater to diverse user expectations, backgrounds, and skill levels. The authors highlight the importance of accessibility, explainability, and short-term engagement, especially during early playing phases. In contrast, inadequate onboarding may disrupt the learning curve and lead to user frustration and premature disengagement [51].

2.5 Explanations in Digital Games

Explanations play an essential role in video game design, shaping how players understand gameplay mechanics, interpret feedback, and engage with both artificial and human opponents. Tutorials are a key method to explain user interactions in games, especially during onboarding [5]. Pagulayan et al. [50] stress that effective tutorials should adapt to players’ skill levels and encourage hands-on learning rather than relying solely on text. Poorly written explanations can frustrate players, whereas well-written explanations do more than just provide instructions; they also direct players toward their objectives. Cao and Liu [11] argue that tutorials must deliver personalized, timely feedback, especially in the context of complex gameplay. Their findings align with broader research emphasizing the variability of optimal tutorial design across genres and demographics [29, 33]. Green et al. [29] advance this by proposing AI-driven tutorials that adapt dynamically to player behavior.

Beyond onboarding, players require clear performance feedback to understand their role in competitive settings. Pfau [53] argues that vague performance feedback fosters overconfidence and toxicity, highlighting the need for explanations that reflect individual contributions in multiplayer environments. In the context of AI in games, explanations help players better understand system behavior. Molineaux et al. [43] present an NPC agent capable of explaining its behavior to players in real time, helping users interpret the rationale behind opaque decision-making. These explanations aim to reduce frustration and foster trust in AI-driven interactions. Similarly, Berger and Müller [6] examine the role of explainable AI in educational games, where transparency is essential for learning or training purposes.

Explanations are not universally beneficial in the same way for all genres and user groups. Rienzo and Cubillos [55] studied games designed for elderly players, emphasizing the need for cognitively accessible instructions. Clear rules and transparent design reduce cognitive load, which can otherwise become a significant barrier to engagement for older adults. On the opposite end of the spectrum, Väkevä et al. [60] examined FromSoftware’s *Dark Souls*, a video game famous for its scarcity of explanations and high difficulty. They argue that the game’s opacity positively contributes to mental resilience by encouraging players to persevere, reflect, and seek support in online communities. Here, the lack of explicit explanations becomes a feature rather than a flaw, providing a high level of difficulty that enables a positive player experience. Denisova et al. [18] argue that repeated failure in games may lead to frustration, but conversely can be considered an essential part of learning if the game provides sufficient explanations.

3 Methodology

3.1 Research Goal and Research Questions

The primary goals of this work are (1) to examine the need for explanations in video games, and (2) to evaluate the effects of explanations on underexplored

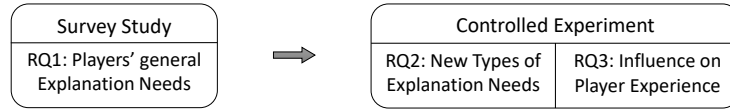


Fig. 1: Overview of the Research Procedure.

UX factors, in the context of a video game. To this end, we conducted a survey and a controlled experiment, addressing the following research questions:

- RQ1** How do players prioritize explanation needs typically found in digital games?
- RQ2** What new types of explanation needs can be identified from users playing the game *Ori and the Blind Forest*?
- RQ3** How do explanations in the game *Ori and the Blind Forest* influence the UX factors fun, immersion, and emotion?

RQ1 explores players' general perceptions of explanations in video games, namely which types of explanations they consider more important than others. We leverage an existing taxonomy for explainability needs in everyday software [21], which digital games are considered to be a part of [28]. Based on this taxonomy, we surveyed video game players to evaluate their perceived need for different types of explanations. RQ2 examines explanation needs that fall outside Droste et al. [21]'s taxonomy. During the experiment, we elicited participants' explanation needs, clustered those that did not align with the taxonomy, and compared them against existing literature. RQ3 addresses a research gap identified by Deters et al. [20], who found that several UX factors are underrepresented in explainability research. Our experiment contributes to this discussion by providing a case example of how explanations in entertainment software can influence fun, immersion, and emotional engagement. Our research procedure is summarized in Figure 1.

3.2 Data Collection

We collected data by conducting two empirical studies:

Survey: The first part of the survey questionnaire consisted of demographic questions, including participants' age, occupation and gaming experience (what and how often they play). The second part of the survey addressed explanation needs in games and built upon the explanation need taxonomy by Droste et al. [21]. In particular, participants used 7-point ordinal scales to gauge the importance of explanations concerning *interactions*, *system behavior*, *domain knowledge*, *privacy & security* and *user interface (UI)*, subdivided into their respective subcategories. The full questionnaire is available in our supplementary material [23].

The survey was published on the online message board of *Leibniz University Hannover* (Germany) in November 2024. From a total of 56 initial survey participants, 50 completed the entire survey. 37 of the participants were students, 16 were employed, and three were occupied otherwise (voluntary service). The average age was 25.3. The majority of participants (33) stated that they rarely

play new games and most (32) stick to one or two genres. Most participants (31) indicated an average play time of less than ten hours per week, with only a few participants (5) stating they play more than 30 hours per week.

Experiment: For the experiment, participants were invited to play the introductory sequence of the video game *Ori and the Blind Forest*. They used either an on-site computer system or *Steam Remote Play* and *Discord*. In both cases, we captured the screen recording and noted participants’ behavior and verbal remarks on an observation sheet. *Ori and the Blind Forest* has an optional setting to deactivate certain interface elements, which includes tutorials and explanations for interface and gameplay elements. We divided the participants of the experiment into two groups: one group playing the game with explanations, and another group playing the game without explanations.

Before and after playing the game, participants answered questionnaires addressing their user experience. The pre-play questionnaire contained the same demographic questions as the initial survey. The post-play questionnaire contained questions to measure three user experience factors relevant to explainable systems and games: fun, immersion, and emotion [20]. In particular, we used questions from the *Game Experience Questionnaire* (GEQ) [34] to measure fun, the *Immersive Experience Questionnaire* (IEQ) [36] to measure immersion, and the *Positive and Negative Affect Schedule* (PANAS) [62] to measure emotion. To ensure that the results of the PANAS questions were not biased by participants’ emotional state before the experiment, we also included PANAS questions in the pre-play questionnaire. These questions were answered on 10-point ordinal scales. Lastly, we included open questions for the elicitation of explanation needs. The full experiment questionnaires are available in our supplementary material [23].

The call for participants was posted on the same message board as the survey, in January 2025. The experiment was conducted with 20 participants, ten of whom were employed, nine of whom were students, and the remaining one was occupied otherwise. Since participation in the preceding survey was anonymous, we do not know if experiment participants had also taken part in the survey. The average age was 25.7. All participants had no prior experience with *Ori and the Blind Forest*, which was a prerequisite for participation. Again, the majority of participants (16) stated that they rarely play new games, and most played games from either one or two genres (8) or three to five genres (9). Reported weekly play time was higher than in the preceding survey, with the majority (11) playing less than ten hours, but four playing more than 30 hours.

3.3 Data Analysis

Survey: The quantitative data from the online survey was analyzed using descriptive statistics. We report participants’ aggregated relevance ratings for Droste et al. [21]’s explanation types, using mean ($\bar{x}$) and standard deviation (σ).

Experiment: The quantitative data from the experiment questionnaires was analyzed using two-tailed hypothesis testing. For all statistical analysis tests presented in this paper, we performed the Mann–Whitney U (MWU) test at the 5% significance level. We evaluated the effect sizes according to Cohen’s

Table 1: Relevance ratings from the initial online survey.
(1 – not at all important, 7 – extremely important)

Type of Need	$\bar{x}$	σ	Type of Need	$\bar{x}$	σ
User Interface	5.04	1.47	System Behavior	4.42	1.87
Domain Knowledge	5.03	1.46	Consequences	4.68	1.68
System-specific Elements	5.60	1.25	Bugs & Crashes	4.60	1.94
Terminology	4.46	1.66	Unexpected System Behavior	4.38	1.99
Interaction	4.97	1.50	Algorithm	4.02	1.86
Navigation	5.22	1.34	Privacy & Security	3.93	1.92
Operation	5.12	1.41	Privacy	3.94	1.97
Tutorials	4.56	1.76	Security	3.92	1.87

guidelines [14]. Participants’ qualitative feedback was categorized using two-cycle coding. To analyze their explanation needs, we performed *in vivo* coding for the first cycle and pattern coding for the second cycle, following the guidelines of Saldaña [56]. During the second coding cycle, we used the coding scheme for explanation needs provided by Droste et al. [21]. The entire coding procedure was conducted by two researchers independently. All conflicts between the two raters were discussed and resolved with the help of a third researcher. All three raters had a background in explainability research. We report the inter-rater agreement using Fleiss’ κ [27]. Codes for newly identified explanation types were declared and assigned by only one researcher, but discussed and reviewed with another.

4 Findings

4.1 Explanation Type Relevance (Survey)

Table 1 shows the survey participants’ aggregated ratings for different explanation types identified by Droste et al. [21]. Explanations concerning the user interface (5.04) or domain knowledge (5.03) received the highest relevance ratings from our participants. In the context of video games, domain knowledge explanations may address elements specific to a given game, such as certain items or characters, or game-specific terminology. User interface explanations concern the heads-up display (HUD) of a game and may address questions about ambiguous design elements or unexpected layout changes. Explanations concerning interactions between the user and the system (4.97) were the next most highly rated explanation types. These explanations may address the operations and navigation that players perform when interacting with a game, but also include tutorials for various gameplay features. The other explanation types, namely those that concerned system behavior (4.42) or privacy and security (3.93), received lower relevance ratings from our participants.

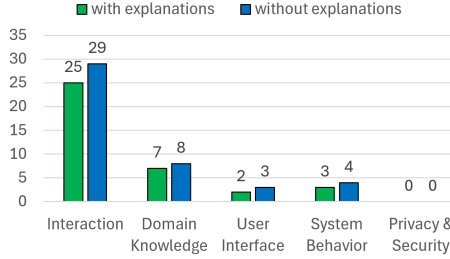


Fig. 2: Explanation needs stated after the experiment (categories by [21]).

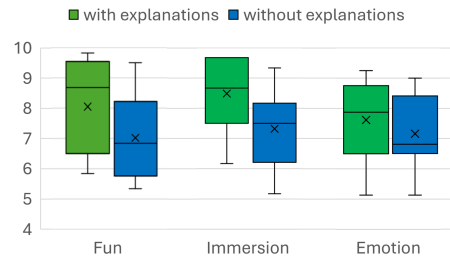


Fig. 3: Players' user experience scores reported after the experiment.

4.2 Identified Explanation Needs (Experiment)

After the experiment, participants stated explanation needs across multiple categories in their answers to the open questions. When coding these needs, we achieved an inter-rater agreement of $\kappa = 0.717$ [27]. According to Landis and Koch [40], this indicates a substantial agreement. Figure 2 shows the number of explanation needs identified in our participants' feedback. The detailed results of our coding procedure are provided in our supplementary material [23].

Known Types of Needs: In contrast to the survey's relevance ratings, needs for interaction explanations were by far the most prevalent during the experiment, followed by explanations for domain knowledge. Only few needs concerned the user interface or system behavior. Notably, participants in the group without explanations stated slightly more needs across almost all categories. The only exception were *Privacy & Security* explanations, for which no need was stated.

New Types of Needs: We observed two major discrepancies between the needs stated by our participants and the need definitions by Droste et al. [21]: (1) Ten explanation needs in the interaction category addressed the navigation of the player character within the game world, rather than the navigation of menus and complex UI elements. These worlds are a key component in many digital games, and are usually not present in other everyday software. (2) Seven explanation needs in the interaction and domain knowledge categories concerned progression within the game in a narrative sense. Specifically, participants stated that they did not understand how to progress the story of the game or what their overall goal in the in-game world was. Outside these categories, two participants explicitly stated that they did not want additional explanations and would rather explore the game themselves. As indicated by our research data [23], their demographic profile cannot be conclusively linked to their preference for no explanations.

4.3 User Experience (Experiment)

Figure 3 summarizes the three user experience factors as reported by our participants after the experiment. In particular, we evaluated whether perceived fun, immersion, and emotion differed between players who received explanations

while playing and those who did not. Players with explanations reported an improved user experience across all three examined factors. The most notable difference was found for immersion, where an MWU test was just short of revealing a statistically significant difference ($p = 0.054, z = 1.928$) at the 5% level. The respective effect size ($r = 0.431$) showed that explanations had a medium effect on immersion [14]. Our evaluation of the next most notable factor, players' perceived fun, revealed no statistically significant difference ($p = 0.097, z = 1.663$) at the 5% level. Similar to our results for immersion, the effect size ($r = 0.372$) showed that explanations had a medium effect on fun [14]. Lastly, we found no statistically significant difference for emotion ($p = 0.308, z = 1.021$) at the 5% level. Furthermore, the respective effect size ($r = 0.228$) showed that explanations only had a small effect on emotion [14].

5 Discussion

5.1 Answering the Research Questions

RQ1) Priority of known Explanation Types: The findings from our survey suggest that players have diverse explanation needs when engaging with the games they play (cf. Table 1). The aggregated scores for most categories were close to the neutral score. This indicates that while some players want explanations, they are not unanimously perceived as a critical component of games. The relative priorities of the explanation types stand in contrast to Droste et al. [21]'s findings for everyday software as a whole. This implies that explainability requirements in games might not be directly transferable from other everyday software domains.

RQ2) Distinct Explanation Needs in Games: Our findings show that players display a distinct need for explanations in games, especially concerning user-system interactions and domain-specific knowledge. The newly identified explanation types concerning software-internal worlds or narrative progression are not a part of Droste et al. [21]'s taxonomy for everyday software, which we attribute to an under-representation of digital games in their dataset.

RQ3) User Experience with(out) Explanations: Within the context of playing the introductory sequence of *Ori and the Blind Forest*, we observed an overall positive effect of explanations on players' user experience. In particular, our findings show notable improvements in player immersion and enjoyment, and a weak improvement in emotional engagement. This highlights the need to explicitly treat explainability as a non-functional requirement in game development, as well-implemented explanations directly influence key user experience factors.

5.2 Reflection on the Results

Distinct Explanation Needs in Games: Throughout our experiment, players who received in-game explanations reported improvements in their user experience.

This highlights a clear demand for explanations that actively contribute to an improved player engagement. Furthermore, the distinct needs and priorities of our participants underline the importance of understanding players as the stakeholders of software explanations.

To assist game developers in creating more effective explanations, future research should explore how different types of explanations specifically influence user experience and overall player engagement. In particular, we suggest the development of a taxonomy for explanation needs in games. The taxonomy used in this paper was designed to work for everyday software systems and has been successfully used for requirements elicitation and management before [21, 46, 47]. However, in the context of games as explainable software, it appears to be limited in its usefulness, as our findings were heavily skewed towards interaction explanations. In contrast to most everyday software, the focus of games lies in the enjoyment of interactions, rather than strictly increasing productivity. This also introduces distinct explanation types, such as narrative explanations, that have not been considered in requirements engineering so far. As such, video games constitute a special case of explainable software, which warrants further examination.

Balancing Challenge and Frustration: Throughout the experiment, two participants stated that they would rather not receive more explanations, even if they were helpful on paper. An essential component of player engagement and flow is the appropriate level of challenge. Determining what constitutes the right set of explanations to challenge players while not frustrating them requires nuanced consideration. This is further complicated by the differences between diverse game genres and player populations. Educational games should prioritize clear, focused explanations and avoid unnecessary challenge, as the goal is to teach or simplify interactions. Conversely, certain genres, such as the FromSoftware’s *Soulslike* games, are renowned for their high difficulty and deliberate withholding of explanations, to cultivate a demanding environment, and foster resilience and perseverance among players.

Future research should investigate the complex interplay between explanations and challenge. Specifically, it should identify when explanations mitigate excessive difficulty and when they diminish intended gameplay tension. Appropriate RE methods can determine the optimal balance of providing explanations that reduce frustration and prevent disengagement, while preserving the level of challenge necessary to sustain flow.

5.3 Threats to Validity

We discuss the limitations of the validity of our research in accordance with Wohlin et al. [63]:

Construct Validity: We constructed our own PX questionnaire, building upon three established sources [34, 36, 62], instead of using the original questionnaires themselves or alternative questionnaires, which might have led to different results.

Considering this limitation, the research reported in this paper should not be considered holistic. Future research should examine the validity of different measures for PX and their usefulness in the context of requirements engineering for explainable systems.

Internal Validity: Our experiment was conducted between-subjects, with a control group. Despite that, all participants played only the introductory sequence of *Ori and the Blind Forest*, instead of more challenging parts that might have led to more or different explanation needs. As such, our findings might not accurately represent how players interact with the game as a whole. Furthermore, our evaluation of PX might not be consistent with the more challenging parts of the game. A within-subjects design, in which the same participant plays different sections of the game with and without explanations, would address these concerns, trading off with the robustness offered by a between-subjects design.

Conclusion Validity: The results from our PX evaluation were not statistically significant and revealed effect sizes from medium to weak. Consequently, they might not be generalizable to a larger population and should be interpreted with caution. In contrast, our coding procedure led to substantial agreement between the raters. Considering the multi-label nature of the procedure, we are confident that our codes are representative of our participants' actual explanation needs.

External Validity: The participants were recruited from within the network of a German university. As such, our findings might not be generalizable to the international gaming community at large. Furthermore, we only experimented with one game, and other games might have led to a different distribution among the identified types of needs. Depending on the specific title and genre, other games might share some of *Ori and the Blind Forest*'s explanation requirements, but might also introduce new ones. The observation that privacy and security explanations received the lowest rating in the survey, and were not at all elicited during the experiment, could be biased by multiple factors. For one, *Ori and the Blind Forest* is a single-player game that requires no personal data from its players. Multiplayer games, where players engage with each other online, might lead to a higher need for privacy explanations. Furthermore, the need for privacy information depends on the specific participant profile, as outlined by Dupree et al. [24]. As the average age of our participants was rather young and recruitment was done in a university context, the low need for privacy explanations could also be influenced by their privacy profile.

6 Conclusion

Video games are complex software and require various explanation types to be accessible to players. Examples of this include in-game tutorials, explanations for operation and navigation, and narrative explanations that build a coherent in-game logic. This research paper addressed games as explainable software, from the perspective of requirements engineering. In particular, we report on a survey and an experiment that examine players' general explanation needs in games and the impact of explanations on players' user experience.

Our findings are twofold: 1) video games invoke distinct explanation needs and priorities, that are not yet considered in requirements engineering for explainable systems, and 2) well-integrated explanations have a positive impact on the UX factors fun and immersion. Consequently, we find that video games constitute a distinct and interesting case of explainable software that should receive special consideration in requirements engineering research.

Data Availability Statement

The responsible university ethics review board approves human-subjects research and they approved this project. All participants have given their informed consent to participate in the studies. To enable the verifiability of our research, we provide our questionnaires (translated to English) and all collected research data (raw) as supplementary material [23]. Qualitative feedback from our participants was anonymized as necessary.

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